

0 4

In fruit flies, males have the sex chromosomes XY and the females have XX.
In fruit flies, a gene for eye colour is carried on the X chromosome. The allele for red eyes, **R**, is dominant to the allele for white eyes, **r**.

0 4 . 1

Male fruit flies are more likely than female fruit flies to have white eyes.

Explain why.

[2 marks]

0 4 . 2

A female fruit fly with white eyes was crossed with a male fruit fly with red eyes to produce a large number of offspring.

Tick (✓) **one** box next to the statement which correctly describes the phenotypes produced from this cross.

[1 mark]

All offspring red-eyed

☐

All females red-eyed, all males white-eyed

☐

All males red-eyed, all females white-eyed

☐

All males white-eyed, females red-eyed and females white-eyed

☐

In fruit flies, the genes for body colour and for wing development are **not** on the sex chromosomes. The allele for grey body colour, **G**, is dominant to the allele for black body colour, **g**. The allele for long wings, **L**, is dominant to the allele for short wings, **l**.

A geneticist carried out a cross between fruit flies with grey bodies and long wings (heterozygous for both genes) and fruit flies with black bodies and short wings.

Table 2 shows the results of this cross.

Table 2

Phenotype of offspring	Number of offspring
Grey body and long wings	223
Black body and short wings	218

0 4 . 3 Explain the results in **Table 2**.

[3 marks]

Question 4 continues on the next page

0 4 . 4

The **first** generation of a population of fruit flies had 50 females.

Calculate how many female fruit flies would be produced from this population in the **fifth** generation.

You can assume:

- each female produces 400 offspring each generation
- half the offspring produced each generation are female
- there is no immigration or emigration
- no flies die before reproducing.

Show your working.

Give your answer in standard form.

[3 marks]

Answer _____

E

